

Research Topics

Federated Data Governance in Single-Enterprise Environments

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1 Introduction

Within large and organizationally complex enterprises, the governance of data has emerged as a strategic design problem that existing organizational structures are often poorly equipped to address. As data volumes grow, the number of systems producing data multiplies, and the analytical demands placed on domain teams intensify, centralized governance structures have proven to be unsuitable for the scale and diversity of modern enterprise data environments. The result is not merely operational inefficiency, but a fundamental misalignment between where data originates and where authority over it resides. This systematic literature review is conducted to establish the conceptual and empirical foundations required for designing a federated, domain-driven data governance framework within a single organization. Section 1.1 defines the key concepts used throughout this study. Section 1.2 provides the organizational and historical context that motivates the research. Section 1.3 describes the emergence of federated governance as the primary structural response to centralized governance failures. Section 1.4 states the research problem that the framework design aims to address.

1.1 Key Concepts and Definitions

Federated Data Governance A federated data governance model is a model in which authority over data assets is distributed across domain-level units that retain meaningful autonomy over locally produced data [15]. The rules for the data assets, i.e. cross-domain standards, policies, and accountability mechanisms, are set by a central governance function. Federation is objectively different from pure decentralization as it preserves a legitimate central authority. The domain-level units hold substantive decision rights over the data they produce. The defining governance challenge of federated models is the design of the authority interface between central and domain-level functions [16, 52].

Decision Rights Decision rights refers to the allocation of authority to make specific categories of decisions about data assets. These include rights to access, create, modify, share, and retire data. They also include the authority to define the standards and policies that govern those actions. Following Weill and Ross (2004) and Fadler and Legner (2022), decision rights in data governance refers to the internal authority over data-related decisions. It does not refer to the legal ownership of data itself [20, 50].

Data Ownership It describes the assignment of accountability and decision control rights for a defined dataset or data product to a specific organizational role. Fadler and Legner (2022) identify three distinct ownership types in big data and analytics contexts: data source ownership, data platform ownership, and data product ownership. Each carries a distinct set of accountabilities that are more complex and more context-dependent than prior research assumed [20].

Domain Autonomy Domain autonomy refers to the degree to which a business domain is authorized to make independent governance decisions about the data it produces and consumes. Domain autonomy is a requirement for domain accountability, because a unit cannot be held responsible for data quality it does not have the authority to control. Autonomy in a federated model is not unconditional, it is exercised within the framework of globally defined standards and is bounded by the central governance function’s authority to set and enforce those standards [7, 34, 36].

1.2 Background and Context

Organizations across every sector are accumulating data at a rate and variety that no previous generation of management had to govern. In addition, the rapid increase of digital applications, connected devices, and automated business processes has transformed data. It went from a by-product of operations into a primary strategic asset that drives forecasting, personalization, compliance, and competitive positioning in their respective markets [20]. Yet, the growing strategic importance of data has not been matched by a corresponding maturity in how organizations govern it. This was documented by Bode et al. (2024) through semi-structured interviews with fifteen industry experts. The study finds that current data architectures are not designed to keep pace with the scale and scope of modern data and analytics use cases, and existing approaches often fail to deliver the business value attributed to them [7].

The consequences of inadequate data governance are not merely operational. Governance failures affect the quality of decisions that depend on data, the ability of organizations to demonstrate regulatory compliance, and the trust on the data products from domain teams and downstream consumers. McCullough (2024) describes enterprise information governance as a strategic framework with the main purpose of assuring accountability in managing decision rights over information and data assets [35]. This framing makes clear that governance is not a technical problem to be solved by better tooling, but an organizational design problem that requires deliberate choices about authority, accountability, and structure. Most of the data governance failures observed in practice are caused by the lack of such deliberate choices or the reliance on governance structures that were designed for a different technological era.

In the past two decades, the primary organizational response to growing data complexity was centralization. Enterprises invested heavily in central data warehouses, data lakes, and master data management platforms. The architectures were based on the assumption that a single, authoritative repository, managed by a central team of data engineers and analysts, could serve as the reliable source of truth for the entire organization. The governance model that accompanied these architectures was similarly centralized. The central data governance team would define standards, manage metadata, enforce quality policies, and act as a referee in disputes about data definitions and ownership.

This model worked when data volumes were manageable, when the number of system producing data was limited, and when the speed of analytical demand was slow. However, as enterprises scaled, the assumptions of centralization broke down on multiple fronts. Butte et al. (2022) observed that the centralized data lake approach is struggling to meet the dynamic business needs of enterprises that were experiencing exponential data growth [47]. Li et al. (2024) argued that the fast increase of data sources, alongside the growing demand for timely analysis and processing, are the fundamental challenges that centralized architectures cannot resolve [21]. The central team had become a bottleneck as it cannot acquire deep knowledge of every domain's data, it cannot respond to every analytical request at the speed the business demands, and it cannot maintain the quality and metadata standards of an increasing number of datasets it does not own and cannot fully understand.

The challenges emerging from the organizational decisions is in fact more problematic than the volume of data. When a central team owns the data that domain teams produce, those domain teams have no accountability for its quality, its fitness for purpose, or its documentation. The producers of data and the people best positioned to understand its meaning, which are the business domains in which the data originates, are structurally disconnected from its governance. The result is what Bode et al. (2024) characterize as a data architecture that fails to deliver on its promise: data that is technically accessible but organizationally ungoverned, accurate by accident rather than by design,

and trusted by no one [7]. Zitoun et al. (2021) developed a data management maturity model for assessing enterprise data capabilities, and they found that most organizations remain at low maturity levels precisely because their governance structures have not kept pace with their data architectures [9]. Therefore, while the technical infrastructure continues to mature, the governance model fails to keep pace.

This finding does not imply that central governance functions are unnecessary. It is that exclusive centralization, i.e. the concentration of all data authority in a team that is structurally remote from the data it governs, is an organizational design that does not scale. Therefore, the challenge is not replacing centralization with its opposite, but to design a governance model that captures the coordination benefits of enterprise-level oversight while distributing data ownership and accountability to where the contextual expertise actually resides.

1.3 Emergence of Federated Data Governance

The most structural response to the limitations of centralized data governance is the data mesh paradigm, developed and described in detail in the book by Dehghani (2022) [16]. The data mesh is a socio-technical approach to analytical data management that rests on four interdependent principles:

1. **Domain Ownership:** The teams that produce data are also responsible for governing it
2. **Data-as-a-Product:** Domain-owned datasets are designed, maintained, and shared as first-class products with defined quality contracts
3. **Self-Serve Infrastructure:** A common platform reduces the technical overhead of data product creation across domains
4. **Federated Computational Governance:** A set of shared standards and policies is enforced across domains without requiring centralized control for every governance decision

Karkošková (2024) brings these principles together by describing the data mesh as a new way to represent the change in thinking about data architecture [25]. Data mesh enables efficient data sharing and creation of data products through a domain-driven approach, product-based thinking, decentralized ownership, and federated governance.

The academic reception of data mesh has grown substantially since 2020. Van Eijk et al. (2024) draw on a systematic review of forty-three gray literature articles and interviews with six data engineering experts [48]. In their paper, they document the architectural design decisions that practitioners face in building the self-serve infrastructure that the data mesh requires. Their finding confirm both the growing real-world adoption of the paradigm and the genuine complexity of its implementation. Kumara et al. (2024) reported on case studies conducted at three organizations in the Netherlands and Germany, the paper provides the first substantial empirical account of what data mesh architectures look like in practice, including the critical design decisions that organizations must make when moving from the theoretical model to deployed systems [28]. Ashraf et al. (2023) draw on the history of microservices adoption to extract lessons learned from data mesh transitions, identifying domain-driven design, team autonomy, and data consistency management as most influential to success [1].

Early empirical evidence supports the proposition that domain-driven data ownership, when successfully implemented, delivers meaningful organizational improvements. Ramchandani and Mackinnon (2023) report on a case study of data mesh adoption in an industrial organization [43]. They find that the establishment of decentralized data ownership and domain-oriented data teams

improved data accessibility, data quality, and the speed of data-driven decision-making. Hooshmand et al. (2022) describe a real-world transformation of a product life cycle management landscape of one of the most data-intensive and organizationally complex environments in manufacturing where the organization transformed from a monolithic, tightly coupled architecture to a federated domain and data mesh [22]. This transformation saw measurable gains in semantic interoperability and in the ability to treat data as a first-class organizational asset.

The data mesh is not merely a technical architecture. According to Wider et al. (2023), the data mesh is a socio-technical approach to decentralized analytics data management [52]. They describe that its success depends as much on the organizational structures, role definitions, and cultural conditions that surround it as well as on the platform infrastructure that enables it. This socio-technical character gives the paradigm both its promise and its difficulty. By aligning data ownership more closely with domain expertise, it addresses the structural failures associated with centralized governance. At the same time however, it introduces a new layer of governance complexity. The principle of federated computational governance requires a central authority that can define and enforce global standards without reverting to the centralized control the paradigm is intended to replace. This raises the question of how the authority structure should be defined, particularly during the transition from centralized to federated governance.

1.4 Problem Statement

The study from Bode et al. (2024), one of the most comprehensive industry study of data mesh implementation to date, find that organizations particularly have difficulty with the transition toward federated data governance [7]. Specifically, organizations struggle to shift the responsibility for data product development and maintenance to domain teams, alongside establishing a comprehensible and operable governance model across the organization. Their recommendation is to introduce a cross-domain steering unit. This idea is practically sensible but theoretically under specified, as it does not address what authority that unit holds, how it relates to domain-level governance structures, or how accountability is distributed between the two. Wider et al. (2023) are more direct about the state of the field. They find a lack of coherence in how data mesh concepts are interpreted in the industry, and almost no rigorous work on how a data mesh platform facilitates governance [52]. Wider, Jarmul and Akhtar (2024) identify data protection compliance as a governance task that is notoriously difficult to automate, and one for which existing data mesh designs offer insufficient specification [2].

The problem becomes more complex with the organizational and cultural dimensions of governance adoption. Nolè et al. (2025) puts emphasis on the technical complexities of implementing a data mesh, which are inseparable from the organizational changes required [38]. Successful adoption demands a fundamental shift in organizational mindset shifting towards taking responsibility for and managing data assets. This is a cultural transformation that most governance frameworks do not design for explicitly. Wang (2025), developed a platform-oriented framework for agile transformation that incorporates federated data governance [49]. They find that delegating data ownership to domain teams while preserving interoperability and global oversight requires explicit architectural choices that bridge local autonomy and enterprise coherence. These choices are currently made inconsistently and ad-hoc without a common theoretical framework.

The problem of data ownership accountability sits at the heart of this challenge. In the study of Fadler and Legner (2022), a four-case study investigation of data ownership in the big data and analytics context, they identify three distinct ownership types: data ownership, data platform ownership, and data product ownership [20]. They show that the accountabilities associated with

each ownership type are more complex and more context-dependent than prior research assumed. Additionally, they find that defining accountabilities in the big data context is more challenging than in traditional operational systems, because data is stored for previously unknown purposes. This underlines why a governance framework that simply assigns ownership labels without specifying the decision rights and accountability mechanisms attached to them is insufficient. Lemieux et al. (2020), examined the governance spectrum between full centralization and full decentralization [29]. They argue that hybrid models, such as federated governance, face distinct challenges that are neither fully resolved by centralization nor by decentralization, and that organizations navigating this transition require deliberate governance design rather than an ad-hoc combination of both approaches.

Pingos et al. (2024) confirm that data meshes introduce challenges related to data ownership, governance, security, monitoring, and observability that have not been resolved by the architectural literature alone [41]. They include findings that reinforce the argument that governance in a federated, domain-driven model requires a dedicated framework treatment rather than a set of architectural patterns.

2 Literature Review

This section explains how the literature review was conducted. The literature review is conducted following the systematic literature review methodology from Kitchenham, 2007 [26].

2.1 Systematic Literature Review (SLR)

The guidelines from Kitchenham are followed for the Systematic Literature Review (SLR) [26]. Kitchenham has defined the main activities for the systematic literature review. These processes can be split up into: Planning, Conducting and Reporting. In the planning phase, the need and understanding of the literature review is defined. The protocol for conducting the literature review is also defined in this section. Then the review is conducted in the conducting phase, where papers are found and filtered based on predetermined criteria. The results will be reported in the reporting phase.

These main processes can be found in Figure 1. The planning will be discussed in section 2.2, followed by how the SLR is conducted in section 2.3 and finally, the systematic literature review with the results in section 3

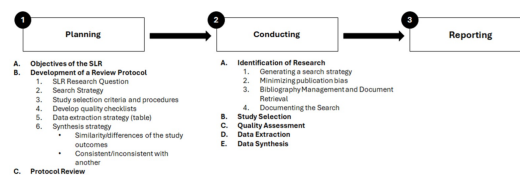


Figure 1: Main steps of the SLR

2.2 Planning

This section outlines the planning phase of the SLR. It is constructed as follows: Section 2.2.1 defines the objectives of the research. Section 2.2.2 defines the research questions for the study. Section 2.2.3 defines the search strategy across different databases, followed by the inclusion and exclusion criteria in section 2.2.4.

2.2.1 Objectives of the SLR

The SLR serves as a basis for the thesis. The framework to be designed for the final project must be include in the full range of what the literature knows about federated data governance. The SLR is conducted to follow the six research objectives, for which each corresponds to the knowledge required for the framework design. The objectives are found below and can be found in the form of a research question in section 2.2.2.

O1 *Identify and analyze the limitations of existing data governance approaches that motivate the adoption of federated data governance within single organizations.*

To understand the need for a new governance framework outside of the centralized ones. The motivation for the research can be found here.

O2 *Identify the theoretical and conceptual foundations through which federated data governance is defined and understood in single-organization contexts.*

How federated data governance is understood in the literature.

O3 *Identify and evaluate the governance constructs proposed or validated in federated data governance frameworks within a single organization context.*

The research aims to map the governance constructs, roles and structural patterns that are aligned with the literature about federated data governance. The result is a taxonomy that will contain the backbone definitions of the framework.

O4 *Identify the organizational, cultural, and change management factors that enable or inhibit the adoption and sustained operation of federated data governance frameworks.*

Failures in adopting new data governance frameworks usually lie outside of technical scope and mostly organizational. Low data literacy, insufficient leadership commitment, domain resistance to accountability obligations, and governance fatigue are recurrent causes of failure in the data governance literature. Here, the goal is to find the assumption or design that a successful federated governance framework needs. The outcome includes the cultural and change management dimensions of the framework artifact, ensuring that the design is grounded in organizational reality rather than technical idealism.

O5 *Identify how existing federated governance models incorporate regulatory compliance and ethical requirements.*

Compliances such as GDPR and EU AI Act need to be adhered to. The result of this objective is to understand the requirements that will put constraints on the framework.

O6 *Identify gaps, unresolved tensions, and design opportunities in the existing literature that justify the need for a novel framework.*

This objective can be treated as the final outcome from the other objectives. The other objectives serve as evidence and this objective will be the outcome of the analysis. The result of this objective is the actionable requirements that are identified before, supporting the need for this thesis.

2.2.2 Research Questions

The six objectives from section 2.2.1 are formulated here as SLR research questions (RQ1–RQ6). Each question is formulated to be answerable from the literature alone, without primary data collection.

- RQ1** *What problem does federated data governance aim to address?*
- RQ2** *How is federated data governance defined, conceptualized, and theorized in a single-organization context?*
- RQ3** *What governance constructs (roles, policies, ownership, accountability, mechanisms) are utilized in federated data governance frameworks?*
- RQ4** *What are the organizational, socio-technical, and implementation challenges for the adoption of federated governance?*
- RQ5** *How are the regulatory and ethical requirements addressed within existing governance models?*
- RQ6** *What conceptual, empirical, and design gaps remain in federated data governance frameworks for single-organization contexts?*

2.2.3 Search Strategy

This section consists of the search strategy used for the literature review. It specifies the databases and search string used. The presented search string is the general search string and database-specific syntax will be applied where required by each platform's query language.

Databases Suitable databases will be selected to collect papers that are relevant for the SLR. The SLR aims to cover literature from information systems governance, organizational science and data engineering. There is not a single database that encapsulates all of the topics, therefore a combination of databases will be used. The databases that will be used are:

D1 Scopus

One of the largest database for IS and CS, covering IS governance, management and data management.

<https://www.scopus.com/search/form.uri?display=advanced>

D2 IEEE Xplore

This is a database for more technical papers encapsulating data engineering, distributed systems and software architecture.

<https://ieeexplore.ieee.org/search/advanced/command>

D3 Web of Science

This database contains high-impact papers, complementing the depth to the breadth of Scopus.

<https://www-webofscience-com.ezproxy2.utwente.nl/wos/woscc/advanced-search>

The gathering of literature will be done using two separate search strings. The first search string is architecture and enterprise context oriented, ensuring identification of structured governance frameworks in hybrid and federated environments. The second search string identifies the organizational and accountability side, finding papers discussing decision rights, domain ownership, and emerging decentralized paradigms. These search strings will be adjusted based on the different databases it is used in depending on their specific syntax requirements.

S1: ("data governance" OR "information governance" OR "governance of data") AND ("federated" OR "shared governance" OR "domain-oriented" OR "domain-driven")

AND

(("hybrid" AND ("cloud" OR "environment")) OR "multi-cloud" OR "federated data" OR "organizational" OR "organisational" OR "enterprise" OR "organizational model") AND
("framework" OR "model" OR "architecture")

S2: ("data governance" OR "information governance" OR "data mesh" OR "data fabric" OR "platform govern*") AND
(federat* OR decentral* OR "domain-driven" OR "domain ownership" OR "data product*" OR "self-serv*") AND
("decision right*" OR accountab* OR "data steward*" OR "data owner*" OR "operating model" OR "governance structure" OR "organizational design" OR "organizational structure" OR interoperab*)

The search strings were developed iteratively through a process of test searches. The strings can be split into three blocks joined by AND operators. Each block is designed to retrieve a predefined condition. Block 1 captures the main domain, i.e. data governance domain, which is then expanded in S2 to include emerging governance paradigms, platform governance and governance discussions that are not labeled as data governance. Block 2 captures the mechanisms of decentralization, where S2 includes wild cards to include different grammatical forms. Block 3 in S1 restricts the results to enterprise-scale hybrid implementations rather than purely conceptual discussions, whereas in S2 it captures governance theories from organizational, IT, and institutional perspectives. Block 4 of S1 ensures that the review includes structured governance designs such as formalized frameworks, conceptual models and enterprise architectures.

Scopus – S1 TITLE-ABS-KEY ("data governance" OR "information governance" OR "governance of data") AND
TITLE-ABS-KEY ("federated" OR "shared governance" OR "domain-oriented" OR "domain-driven") AND
TITLE-ABS-KEY (("hybrid" AND ("cloud" OR "environment")) OR "multi-cloud" OR "federated data" OR "organizational" OR "organisational" OR "enterprise" OR "organizational model" OR "organisational model") AND
TITLE-ABS-KEY ("framework" OR "model" OR "architecture")

Scopus – S2 TITLE-ABS-KEY ("data governance" OR "information governance" OR "data mesh" OR "data fabric" OR "platform govern*") AND
TITLE-ABS-KEY (federat* OR decentral* OR "domain-driven" OR "domain ownership" OR "data product*" OR "self-serv*") AND
TITLE-ABS-KEY ("decision right*" OR accountab* OR "data steward*" OR "data owner*" OR "operating model" OR "governance structure" OR "organizational design" OR "organizational structure" OR interoperab*)

IEEE Xplore – S1 ("Data Governance" OR "Information Governance" OR "governance of data") AND
("federated" OR "shared governance" OR "domain-oriented" OR "domain-driven") AND
(("hybrid" AND ("cloud" OR "environment")) OR "multi-cloud" OR "federated data" OR "organizational" OR "organisational" OR "enterprise" OR "organizational model") AND
("framework" OR "model" OR "architecture")

IEEE Xplore – S2 ("data governance" OR "information governance" OR "data mesh" OR "data fabric" OR "platform govern*") AND
 (federat* OR decentral* OR "domain-driven" OR "domain ownership" OR "data product*" OR "self-serv*") AND
 ("decision right*" OR accountab* OR "data steward*" OR "data owner*" OR "operating model" OR "governance structure" OR "organizational design" OR "organizational structure" OR interoperab*)

Web of Science – S1 TS=("Data Governance" OR "Information Governance" OR "governance of data") AND
 TS=("federated" OR "shared governance" OR "domain-oriented" OR "domain-driven") AND
 TS=(("hybrid" AND ("cloud" OR "environment")) OR "multi-cloud" OR "federated data" OR "organizational" OR "organisational" OR "enterprise" OR "organizational model") AND
 TS=("framework" OR "model" OR "architecture")

Web of Science – S2 TS=("data governance" OR "information governance" OR "data mesh" OR "data fabric" OR "platform govern*") AND
 TS=(federat* OR decentral* OR "domain-driven" OR "domain ownership" OR "data product*" OR "self-serv*") AND
 TS=("decision right*" OR accountab* OR "data steward*" OR "data owner*" OR "operating model" OR "governance structure" OR "organizational design" OR "organizational structure" OR interoperab*)

2.2.4 Inclusion and Exclusion Criteria

This section explains how the collected papers were screened and selected. The inclusion and exclusion criteria establish the scope of the review by defining what constitutes acceptable evidence and what falls outside its boundaries. These criteria were determined before the search was conducted. In accordance with Kitchenham [26], the inclusion and exclusion criteria should be developed based on the research question. They are then applied to identify and filter relevant studies by specifying the characteristics that each study must meet to be included in the review. The paper must satisfy all inclusion criteria to be included and excluded when any of the exclusion criteria is met. Alongside their aspect, the inclusion and exclusion criteria are stated below.

IC1 Papers addresses data governance, information governance, or data management as one of its main focus

IC2 Paper discusses the governance structures; federated, decentralized, domain-driven or hybrid

IC3 Paper addresses governance within an organizational context. Inter-organizational and public sector papers are eligible where their governance design principles are transferable to intra-organizational federated structures.

IC4 Papers discussing data ownership models

IC5 Paper addresses governance mechanisms affecting authority or accountability alignment

IC6 Papers published from 2015 to the present serve as the primary review window. Seminal pre-2015 works, including foundational literature published before 2010, are included where they provide essential conceptual or theoretical grounding.

EC1 No Governance framework/architecture or mentioned only once

EC2 Pure technical, i.e. distributed systems or IT governance without data governance

EC3 AI Governance only

EC4 The full text is not available or accessible online

EC5 The study is in a different language than English or of low quality

2.2.5 Quality Checklists

Kitchenham emphasizes the importance of a quality check on the collected studies [26]. The quality assessment checklist in Table 1, contains quality questions are defined to assess the quality of the literature. A three-point scale is applied where: 0 (criterion not met), 0.5 (criterion partially met), or 1 (criterion fully met). Thereafter, the scores will be summed for each study with a maximum of 10 for a single study. This enables the sorting of the literature based on their paper quality. Furthermore, the paper will be excluded if the summed score of a single paper is less than 6. This represents the minimum at which a study can be considered methodologically sufficient across the majority of criteria. This adds another filter criteria, ensuring the papers with low quality are excluded.

#	Quality Assessment Item	Scoring
QA1	Is the research question or design objective clearly and explicitly stated?	0 / 0.5 / 1
QA2	Is the theoretical or conceptual grounding of the study described?	0 / 0.5 / 1
QA3	Is the methodology or design approach described to permit replication or evaluation?	0 / 0.5 / 1
QA4	Are the data sources or evidence base identified?	0 / 0.5 / 1
QA5	Are the findings or framework components directly traceable to the evidence or argumentation presented?	0 / 0.5 / 1
QA6	Are the limitations of the study explicitly acknowledged?	0 / 0.5 / 1
QA7	Does the paper address governance structures (roles, accountability, decision rights) rather than exclusively technical systems?	0 / 0.5 / 1
QA8	Is the organizational or research context described?	0 / 0.5 / 1
QA9	Are the core constructs (decision rights, data ownership, accountability) utilized in practice?	0 / 0.5 / 1
QA10	Do the conclusions extend beyond the immediate research context in a way that contributes to generalizable knowledge?	0 / 0.5 / 1

Table 1: Quality Assessment Criteria

2.2.6 Data Extraction Strategy

To manage all of the literature collected from the search string, the free, open-source reference manager Zotero is used. The reference manager assists in saving links and abstracts of the literature collected from the databases. In addition, the reference manager assist in generating bibliographies and saving citations to be utilized in this report. The screening of the literature's title and abstract is executed using Covidence and an export from the Zotero collection. The quality assessment was

done in Excel for a proper overview of what each literature exactly entails. The full assessment can be found in Appendix.

For the final papers, a standardized data extraction form developed in advance of the review will be used. Standardized extraction is required to ensure that the data collected from each paper is consistent in type and depth. In addition, it ensures that the extraction process is replicable, and that the resulting data is in a form suitable for thematic synthesis [26].

The extraction form captures the following fields for each paper:

- (1) Bibliographic details (author, year, title, source, DOI)
- (2) Publication type and quality score
- (3) RQ relevance (RQ1–RQ6) and secondary relevance where applicable
- (4) Research approach and methodology
- (5) Organisational context (sector, size, country where reported)
- (6) Governance constructs addressed and how they are operationally defined
- (7) Key findings or design contributions relevant to each RQ
- (8) Limitations stated by the authors
- (9) Citations to other corpus papers (for within-corpus citation analysis)
- (10) Reviewer notes on the paper’s contribution to the gap analysis (RQ6)

2.2.7 Synthesis Strategy

Summarizing the data is done with the synthesis strategy following from Kitchenham (2007)[26]. After extracting the data using the data extraction form from section 2.2.6, a synthesis strategy can be defined to formulate the results from the SLR. For this SLR, the descriptive synthesis is used. Descriptive synthesis is the process of organizing and summarizing literature to show what is known, how studies relate to one another, and where important gaps remain.

2.3 Conducting

This section will provide information on the steps taken when conducting the literature review, including statistical results on quality mentioned in section 2.2.5. Section 2.3.1 shows the PRISMA flow of the collected studies. It ends with the resulting number of literature.

2.3.1 Literature Screening

Database	Literatures from S1	Literature from S2
Scopus	53	190
IEEE Xplore	45	170
Web of Science	25	99

Table 2: This table shows the number of literature obtained from the different databases using S1 and S2

In addition to the journal articles, several scholarly books were identified within the collection. These works provide the foundational framework for the study, establishing its conceptual origins and situating the research within the academic field. The five following books serve as the foundation for this study's theory:

- Weill & Ross (2004) — IT Governance: How Top Performers Manage IT Decision Rights for Superior Results [50]
- Dehghani, Z. (2022) — Data Mesh: Delivering Data-Driven Value at Scale [16]
- Majchrzak et al. (2023) — Data Mesh in Action [34]
- Menon, P (2024) — Data Mesh: Principles, Patterns, Architecture, and Strategies for Data-Driven Decision Making [36]
- Curry et al. (2022) — Data Spaces: Design, Deployment and Future Directions [13]

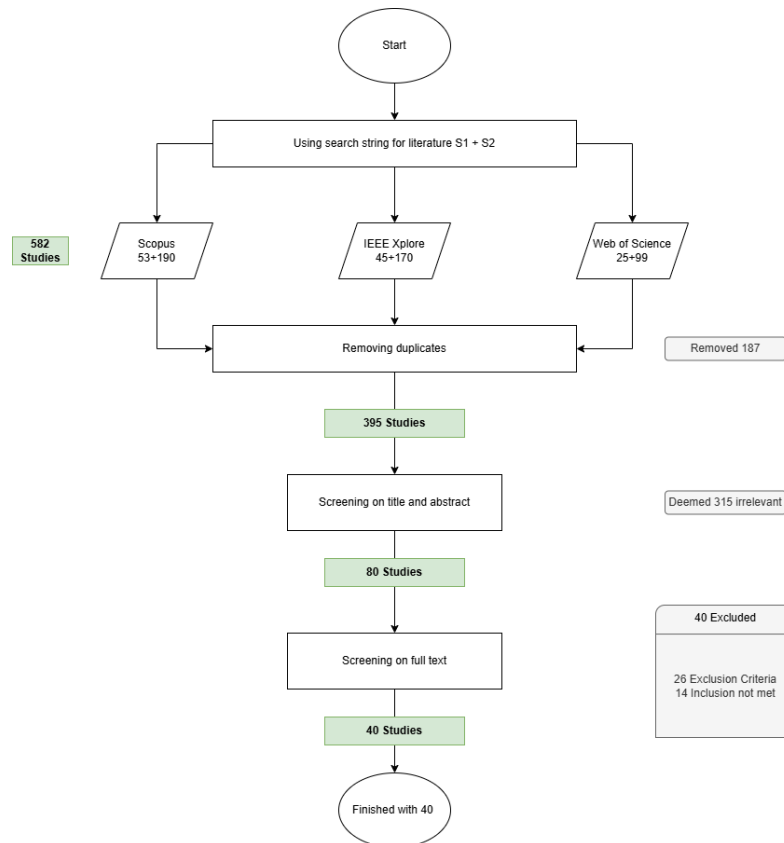


Figure 2: Prisma flow diagram for this systematic literature review

The total number of literature fetched from the databases was 582. The large number of literature is the result of search string 2 (S2), which was a broad search on data governance. After filtering the overlapping documents from the different databases, the resulting number of unique studies was 395. On these 395 studies, the inclusion and exclusion criteria was applied on the title and abstract of each literature. During these phase, the excluded literature falls in the following categories

- Lack of governance, aligning with exclusion criteria 1. Number of literature excluded: **262**

- Technical governance or no data governance, aligning with exclusion criteria 2. Number of literature excluded: **50**
- AI Governance only. Number of literature excluded: **2**
- Pre-2009 and incorrect domain. Number of literature excluded: **1**

This has reduced the number of literature that deem relevant to 80. The next phase of the screening is analyzing its full text, understanding whether the literature has something to offer for this research. The full text analysis will follow the same inclusion and exclusion criteria defined in 2.2.4. During the full-text analysis, several studies were excluded for specific reasons. The most common exclusion category was EC1, these papers are technical in nature and where governance aspects were insufficient. These papers focused mainly on topics such as CNN–RNN policy enforcement, blockchain infrastructure or machine learning–based security systems that lacked federated or domain-driven governance elements. The second most frequent reason was EC2, covering papers that addressed governance in an inter-organizational or cross-national context without transferrable principles that would contribute to this study. In addition, one paper was excluded under EC4 because the full text was only available in Chinese. This has led to the final number of 40 papers left for the study. The full screening process can be visualized using PRISMA [39]. The PRISMA visualization can be found in figure 2.

2.3.2 Quality Assessment of Literature

In this section, the quality of the literature is analyzed. This was done based on the quality assessment questions created in Section 2.2.5. The points awarded for each literature selected from the screening will be briefly explained in this section. As mentioned in section 2.2.5, the maximum score for a single paper is 10, and studies that score less than 6 points are excluded from the study. The top 10 papers from the literature and their scores can be found in Table 3. The full detailed analysis of each paper can be found in Appendix A.

#	Paper / Authors (Year)	Type	Total
1	Bode et al. (2024) [7]	J	10.0
2	Kovach et al. (2025) [27]	J	10.0
3	Nguyen-Phan et al. (2025) [37]	C	9.5
4	Parmiggiani & Amagyei (2025) [40]	J	9.5
5	Garcia et al. (2022) [42]	J	9.5
6	Lemieux et al. (2020) [29]	J	9.5
7	Fadler & Legner (2022) [20]	J	9.5
8	Weill & Ross (2004) [50]	B	9.5
9	Mendonça et al. (2025) [19]	C	9.0
10	McCullough (2024) [35]	J	9.0

Table 3: Top 10 included papers by quality assessment score (full 10-point scale). Type codes: *J* = peer-reviewed journal; *C* = conference paper; *B* = book/monograph.

Note: Six papers share a total of 9.5 at positions 3–8 and two share 9.0 at positions 9–10. Within tied groups, papers are ordered by relevance to the main research question.

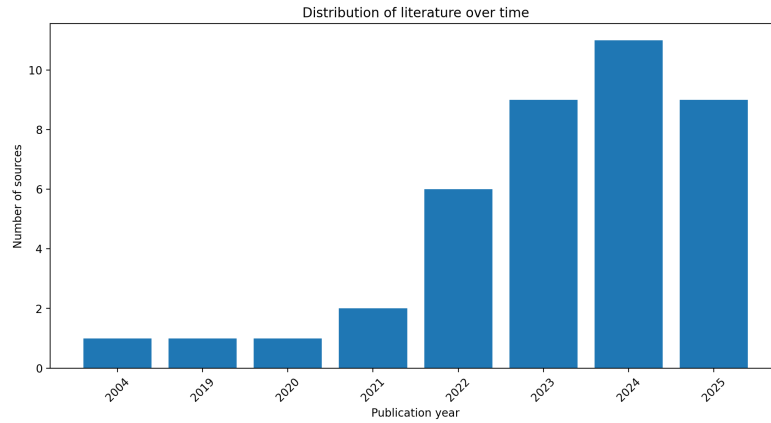


Figure 3: Distribution of literature over time

3 Literature Review Results

This section presents the findings of the systematic literature review through a thematic synthesis of the selected studies. The reviewed literature is examined to identify recurring patterns, conceptual tensions, and substantive contributions across the papers. Section 3.1 maps the broader landscape of the literature and identifies cross-cutting themes that inform the subsequent analysis. Section 3.2 addresses each of the research questions defined in Section 2.2.1 and synthesizes the literature accordingly.

3.1 Literature Overview

The 40 papers span over 2004–2025, with the majority of publications appearing between 2022 and 2025, this can be seen in Figure 3. This concentration aligns with the period in which data mesh emerged as a widely discussed paradigm for enterprise data architecture, alongside growing academic interest in federated governance and data spaces [7, 16, 30, 36]. The papers are predominantly composed of IEEE conference papers and journal articles (approximately 55%), followed by Elsevier, Springer, and MDPI journal contributions (approximately 25%), workshop and short papers (approximately 15%), and practitioner-oriented books (approximately 5%). This distribution reflects the sampling choices made during the search process, and should not be taken as a definitive characterization of the field’s disciplinary balance.

Geographically, the papers mainly originate from Europe (Germany, the Netherlands, Switzerland, France, Spain, and the United Kingdom), with notable contributions from North America (the United States and Canada), Asia (China, Singapore, and Taiwan), and South America (Brazil and Colombia). The European concentration is consistent with the regulatory context that motivates much of this work, including the General Data Protection Regulation (GDPR) and the EU Data Governance Act, as well as the GAIA-X initiative for European data infrastructure [13, 19]. The global distribution reflects a genuinely cross-regional academic interest in federated governance models and distributed data architectures.

3.1.1 Cross-Cutting Themes

Several themes consistently emerge across the reviewed literature. The following section synthesizes these recurring patterns to highlight the key concepts and perspectives that shape the current understanding of the topic.

The most common tension in the corpus is that between centralization and decentralization in governance authority [7, 16, 34, 52]. Most papers do not treat this as a binary choice, but as a design problem that requires a structured balance between globally enforced standards and local autonomy in implementation. Within the data mesh literature specifically, the self-serve data platform is proposed as the primary mechanism through which this balance is operationalized, enabling domain-level autonomy within centrally managed policy constraints [48]. This framing does not, however, apply uniformly across the literature, as papers in the blockchain governance and data spaces traditions approach the same tension through different architectural means, including smart contracts, decentralized identifiers, and federated trust frameworks [30, 55].

A second, recurring observation concerns the socio-technical character of data governance [16]. Although many papers argue that the organizational and cultural dimensions of governance are just as important as the technical ones, the distribution of research attention does not reflect this balance. Technical contributions, such as architectural patterns, platform designs, and enforcement mechanisms, substantially outnumber organizational or empirical studies of governance adoption and failure. This imbalance is itself a finding of relevance to the present research, as it suggests that existing frameworks may be structurally under specified with respect to the organizational conditions they assume. Bode et al. represent a notable exception, identifying challenges such as responsibility shift, domain resistance, resource constraints, and conceptual misappropriation as recurrent impediments to federated governance adoption [7].

Finally, the papers provide an early-stage engagement with AI-assisted governance, centered on the potential for large language models (LLMs) and automated agents to support policy verification, compliance checking, and access control [2, 3]. This work builds on the more established concept of computational data governance, in which governance tasks are automated through platform-level tooling [16, 52]. The extension into generative AI remains novel and largely conceptual in character. The empirical and conceptual foundations for LLM-based governance assistance have not yet been rigorously established in the literature [3]. This constitutes an emerging direction rather than an established finding and is outside of the scope of this research.

3.2 Thematic Synthesis by Research Question

This section synthesizes the selected studies in relation to the research questions guiding the literature review. Using the research questions, this section highlights the key ideas behind federated data governance, its challenges, and gaps across the literature.

3.2.1 The Problem Addressed by FDG

The problem that federated data governance is designed to address is not primarily one of data volume or technological complexity, but of governance authority and structural design [16, 50]. Within single organizations, centralized governance arrangements are met with three compounding structural failures. First, they produce a knowledge-locality failure where governance teams hold formal authority but lack the domain expertise needed to exercise it effectively. This results in decisions that may be technically feasible yet operationally detached from the data they govern [7, 52]. Secondly, the centralized governance has trouble scaling as the number of domains, data

products, and use cases grows, the central governance function becomes a bottleneck. The bottleneck causes data requests to accumulate, pipeline maintenance to overwhelm analytical work, and governance quality to deteriorate under volume [16]. Thirdly, centralized governance introduces accountability diffusion. When governance responsibility is located in a central team, domain teams have little incentive to take ownership of data quality or compliance. As a result, responsibility for failures becomes ambiguous [7, 20].

Federated governance addresses these three failures by redistributing governance authority to the domains that produce and use data, rather than merely redistributing governance tasks. At the same time, it preserves global standards through a shared self-serve platform and a coordinating steering body [16, 48, 52]. It is equally important to note that this redistribution is not solely a technical arrangement. Lemieux et al. demonstrate that organizations operating in hybrid states frequently show issues from both centralized and decentralized models simultaneously. This is especially the case when governance authority is only partially decentralized and the underlying authority structures remain unresolved. [29]. This finding clarifies the core design challenge: a viable federated data governance framework must establish a coherent authority structure rather than simply provide a set of governance tools. It must also be capable of functioning coherently during organizational transition, not only in a fully mature steady state.

3.2.2 Conceptualization and Theoretical Foundations of FDG

The reviewed literature draws on several distinct theoretical and conceptual frameworks to define and understand federated data governance. Together, these frameworks address RQ2 from complementary angles, covering governance operating models, structural theory, definitional precision, and accountability taxonomy. None was developed exclusively for single-organization contexts, but each provides transferable conceptual resources.

The most common framework from the literature is federated computational governance, as formulated within the data mesh paradigm [16, 34, 52]. Under this model, governance is defined as a decision-making structure led by a federation of domain data product owners and data platform owners who maintain local autonomy while collectively adhering to a set of globally defined rules [16]. These global rules address cross-cutting concerns such as security, privacy, and interoperability, and are informed by domain-specialist input from legal, compliance, and platform stakeholders. The distinguishing feature of this model is its computational character. Governance policies are not enforced through manual or bureaucratic processes. Instead, they are automated at the level of each data product through a shared self-serve infrastructure platform [16, 34]. Majchrzak et al. define federated computational data governance as a data governance model in which policies are translated into algorithms and enforced automatically by the IT infrastructure wherever feasible [34]. This operational definition is the most widely represented framework in the papers, appearing across the majority of data mesh oriented papers either as a primary framework or as an explicit point of reference.

A second theoretical lens is polycentric governance, applied to data infrastructure contexts by Parmiggiani and Amagyei [40]. Drawing on Östrom’s foundational work, polycentric governance describes a decision-making system in which multiple semi-independent centers of authority coexist and interact within a broader framework of shared rules and norms. Rather than being subordinated to a single hierarchical authority, these centers operate in relation to one another within that shared framework [40]. Parmiggiani and Amagyei apply this lens to a distributed scientific data infrastructure, demonstrating how local actors manage data as a shared resource through collaborative curation practices that both enact and inform top-level governance rules. It should be noted

that their empirical context is cross-institutional rather than intra-organizational; the theoretical transfer to single-organization settings requires treating organizational domains as the relevant semi-autonomous units, and treating the platform and governance council as the overarching framework of shared rules. With this adaptation, polycentric governance offers a theoretically grounded account of how domain autonomy and global coordination can coexist without collapsing into either hierarchy or fragmentation. Aguerre’s application of polycentrism to global internet data governance addresses the inter-national and multi-stakeholder level. However, it reinforces the broader finding that polycentric arrangements are characterized by overlapping mandates and the absence of a single final arbiter of authority. Coordination in such arrangements is achieved through norms and practices rather than through centralized command. [4].

McCullough contributes a definitional clarification for federated data governance in a single organization context [35]. He defines enterprise information governance as a corporate-wide strategic framework that uses control mechanisms to ensure accountability in the management of decision rights over data assets. He explicitly distinguishes it from data management, which is the operational execution of data tasks. This distinction is important as federated governance operates at the level of authority structures and accountability allocation, not at the level of operational data handling. McCullough further situates enterprise information governance within a target operating model, positioning it as a form of business architecture. For the purposes of this research, this framing underlines that a federated governance framework is primarily an organizational design problem, not a technical one.

Finally, Fadler and Legner contribute a taxonomic extension that grounds governance accountability in the operational roles created by big data and analytics environments [20]. Through a multi-case study, they identify three distinct ownership types:

1. **Data ownership:** Accountable for source data supply, integrity, and lifecycle
2. **Data platform ownership:** Accountable for mediating data flows across the infrastructure and managing dependencies between producers and consumers
3. **Data product ownership:** Accountable for the analytical value and lifecycle of domain-level data products

These three types are not merely descriptive categories; they represent structurally distinct governance roles that arise from the decoupling of data supply, platform mediation, and data demand that characterizes modern analytical architectures. The data platform owner in particular occupies a coordination role that traditional governance frameworks, which focus primarily on source data stewardship, do not account for. Fadler and Legner argue that data governance designs must be extended to include these analytics-related accountabilities, which aligns directly with the federated governance model’s requirement to distribute responsibility across domains while preserving system-wide coherence.

3.2.3 Governance Constructs

The papers discuss multiple different governance constructs applicable for federated data governance. These constructs can be split into five different clusters:

- Policy and compliance mechanisms
- Ownership and accountability structures

- Data contracts
- Privacy and access control
- Democratic and cooperative mechanisms

While each cluster addresses a distinct governance concern, they are not mutually exclusive. In practice, their implementations require constructs from multiple clusters simultaneously.

The most commonly discussed area focuses on policy and compliance mechanisms, with policy-as-code emerging as the dominant paradigm across the papers. In this approach, governance rules are expressed in machine-executable form rather than as natural-language documents, enabling automated, consistent enforcement at runtime [32, 52]. Open Policy Agent (OPA) is identified across multiple sources as the current implementation tool, with governance logic declared in the Rego language and evaluated at Policy Decision Points (PDPs). A notable architectural contribution by Brambilla and Plebani addresses the scalability of this model through PDP replication strategies, thereby bridging policy coherence and enforcement performance in distributed data environments [32]. Future development of this cluster involves AI-assistance at its core, utilizing LLMs to convert natural language regulations into data contracts.

The second cluster concerns ownership and accountability, which is the foundational organizational construct of federated governance. In the papers, data ownership is described as domain-level accountability with responsibilities for data product creation, quality assurance, and lifecycle management [7, 20, 48]. Fadler and Legner’s typology of ownership roles provides details at an operational level that distinguishes between different forms of data accountability, while Bode et al. introduce the cross-domain steering unit as an enterprise-level coordination body capable of exercising governance authority without reverting to the centralization that federated architectures seek to avoid. Together, they establish a layered accountability model that spans both local domain and global enterprise levels.

Data contracts are the third and increasingly important governance cluster. Functioning as formal artifacts that define the terms of data exchange between producers and consumers, data contracts specify schema obligations, service-level agreements (SLAs), and access constraints [2, 49, 52]. Their importance lies in the fact that they make governance obligations explicit at the point of data sharing, rather than leaving them to implicit conventions or centralized enforcement after the fact. Within a single organization, data contracts function less as negotiated agreements between independent parties and more as structured declarations of how a data product will behave and what conditions govern its use. They operationalize federated governance at the data product level by giving domain teams and consumers a shared, traceable record of the standards that apply to a given data exchange [3, 52].

A fourth cluster addresses privacy and access control. The papers contribute a range of technically sophisticated mechanisms adapted to federated environments. These include proxy re-encryption for privacy-preserving data delegation, attribute-based encryption with hidden policy disclosure, the PROTON privacy-by-design framework for data products, and GDPR-aligned consent management architectures [42, 44, 55]. While these mechanisms vary in their technical approach, they share a common concern with enforcing access control and privacy guarantees in settings where data crosses domain boundaries, a challenge for federated architectures.

The fifth and most novel cluster in the context of corporate data governance is the democratic and cooperative mechanisms. Mendonça et al. and Bühler et al. introduce governance constructs from collective decision-making theory [8, 19]. These constructs include voting mechanisms, cooperative principles, and participatory governance structures. These mechanisms are underrepresented in

the existing data mesh literature, which has historically prioritized technical development and enforcement over participatory governance design. These papers suggest a growing recognition that governance is not only a technical issue, but also one of legitimacy and domain-level acceptance, and that ideas from related fields, such as cooperative economics and polycentric governance, may be valuable in federated organizational contexts.

3.2.4 Organizational, Socio-Technical, and Implementation Challenges

The transition to federated data governance is not primarily a technical problem. The literature identifies a cluster of organizational, cultural, and structural challenges that shape whether the transition succeeds or fails. These challenges can be grouped into four broad categories: cultural resistance, incentive misalignment, comprehension gaps, and structural tensions between autonomy and standardization.

Firstly, cultural resistance and organizational silos remain a major challenge in organizations attempting to implement federated data governance. Many organizations attempting federated data governance inherit cultures in which data is treated as a departmental asset rather than a shared resource [7, 16]. Departments protect their data because sharing it is seen as a loss of advantage. Breaking down these silos requires a cultural transformation that extends well beyond IT. It must reach product, management, and sales teams as well [34]. Dehghani describes this as a shift from treating data as a by-product of operations to treating it as a product that is actively managed and shared [16]. In practice, this shift is slow. Bode et al. find that acceptance and resistance are among the most consistently reported challenges across different organizational types, and that they persist even after technical infrastructure is in place [7].

Secondly, incentive misalignment and the extra work mentality represent a major barrier to federated data governance. Federated data governance delegates data product ownership to domain teams, whose primary focus is typically operational rather than analytical [7]. Domain teams often experience the documentation, maintenance, and provisioning of data products as extra work that takes time away from their core responsibilities. Bode et al. report that domain teams frequently receive no direct compensation for the effort they put into providing data for other domains, which leads them to lower the priority on these tasks [7]. This problem is worsened when organizational reward systems are based entirely on local business unit performance rather than on enterprise-wide data sharing outcomes. Weill and Ross note that governance arrangements consistently perform worse when incentive systems pull in the opposite direction [50]. Dehghani argues that the solution requires both intrinsic motivators, such as autonomy and purpose, and extrinsic incentives that link team rewards to the success and adoption of shared data products [16].

Thirdly, comprehension gaps and limited data literacy are recurring barriers to federated data governance. A common issue is the shallow understanding of what federated data governance actually requires. Bode et al. find that employees and managers often adopt new terminology without making any real structural or technical changes [7]. For instance, interviewees describe cases where a centralized data warehouse was simply renamed a data mesh without any redistribution of ownership or automation of governance. This misinterpretation blocks progress, because governance transitions that rely on surface-level understanding fail to produce the domain accountability that makes federation work. Sustained operation requires broad data literacy across domains, ongoing training, and clear communication about what the governance model actually demands from domain teams [7, 34].

Fourthly, change management and structural support are essential to the implementation of federated data governance. Because federated data governance involves reorganizing teams and

redistributing decision-making power, which produces friction and internal resistance [7, 34]. Effective change management is a key enabler. Bode et al. recommend pursuing small, low-cost, and visible *quick wins* in the early phases of adoption [7]. These wins build organizational momentum and demonstrate value to senior management before the full costs of transformation become visible. In addition, a cross-domain steering unit is recommended to coordinate governance policy, prioritize use cases, and enforce global standards during the period when full automation is not yet available [7]. Executive sponsorship is necessary to sustain the transition, as organizational frameworks in established companies are often inflexible and resistant to changes in financial resource allocation [7].

Fifthly, the tension between autonomy and standardization represents a deeper socio-technical challenge in federated data governance. At its core, this challenge concerns the balance between domain autonomy and global interoperability. If global standards are enforced too rigidly, domain teams lose the flexibility to manage their data in ways that reflect local knowledge and business context [19, 52]. If standards are too loose, the result is disconnected data products that cannot be combined across domains [16]. Mendonça et al. describe this as the standardization-flexibility dilemma where interoperability requires alignment with shared schemas and protocols, but standardization constrains local innovation [19]. A related problem is what they call the automation-agency paradox. Excessive automation of compliance policies removes human judgment from governance decisions, which can reduce the sense of accountability among domain teams. Insufficient automation creates manual bottlenecks that prevent the governance model from scaling [19]. Neither extreme is workable, and the appropriate balance depends on organizational context. This balance is further complicated by power asymmetries between domains. Domains with more resources tend to have greater influence over shared governance standards and platform design. As a result, the resulting policies may not properly reflect the data quality requirements or operational constraints of smaller or less experienced domains.[19]. The automation level that seems workable to a well-resourced domain may be unrealistic for a domain with fewer technical capabilities, yet both are subject to the same global governance rules.

Finally, Resource and capability constraints represent a major practical barrier to the adoption of federated governance. Restructuring the landscape of legacy systems and creating independent data pipelines for each domain is costly in time, money, and specialized skill [7, 34]. Organizations face a skills gap because building and maintaining data products at scale requires engineers with data engineering expertise distributed across every business domain, not concentrated in a central team. Bode et al. find that resource limitations are one of the most consistently reported practical barriers to adoption, particularly in established organizations with large legacy infrastructures [7]. The costs are difficult to justify internally, partly because the financial benefits of improved data quality and accessibility are hard to quantify in concrete business terms.

Taken together, these findings show that the organizational and cultural dimensions of federated data governance are more difficult to manage than the technical ones. The literature broadly agrees that governance transitions require cultural change, aligned incentives, data literacy across the organization, structural support from leadership, and a carefully calibrated balance between global standards and local autonomy. A federated data governance framework that is technically sound but organizationally unsupported is unlikely to sustain itself in practice [7, 16].

3.2.5 Regulatory and Ethical Requirements in FDG

Existing federated governance models handle regulatory and ethical requirements through two distinct but complementary mechanisms: the structural embedding of compliance obligations into

governance architecture, and the computational automation of policy enforcement at the data product level. The papers indicate that the regulatory dimension is more developed than the ethical dimension, and that significant implementation gaps remain in practice even where conceptual solutions exist.

The most common regulatory context across the papers is EU personal data protection law, particularly the General Data Protection Regulation (GDPR). Federated computational governance, as defined by Dehghani, explicitly addresses GDPR requirements. This is done by treating regulatory obligations as global policies. These policies are codified and enforced at the platform level rather than managed manually by individual domain teams [16]. The GDPR’s right to erasure, for instance, is identified as a cross-domain governance concern. How and when this right is triggered is defined globally. The same applies to how each data product assures its execution. These requirements are then delegated computationally to the platform [16]. Similarly, PII protections such as access control, data localization, consent tracking, and retention limits are implemented as policy-as-code, enabling domain teams to declare policies in machine-readable form while the platform enforces them automatically [16, 52].

The practical challenge of utilizing this approach is significant. Wider et al. demonstrate that while data contracts provide a structural mechanism for encoding access constraints and data usage conditions, the upstream translation of legal documents into formal contract specifications is tedious and error-prone, and must be repeated whenever regulatory requirements change [52]. To address this, Wider, Harrer, and Dietz propose Governance AI, a large language model based tool that evaluates data access requests directly against plain-text privacy policies, removing the requirement for prior translation into formal rule languages [3]. This represents an emerging direction toward more adaptive regulatory compliance, though the authors acknowledge that the approach requires empirical validation and should not fully replace human expert oversight.

The EU Data Governance Act adds a distinct layer of structural requirements for federated governance, particularly its neutrality and organizational separation mandates for data intermediaries [19]. Mendonça et al. map the DGA’s Article 11 requirements to cooperative governance mechanisms, demonstrating that automated compliance monitoring and conflict-of-interest detection can be used to satisfy regulatory reporting requirements [19]. Liang et al. identify cross-border data governance as a further structural challenge, noting that tensions between GDPR adequacy requirements and data localization rules in other jurisdictions produce compliance fragmentation that no single automated governance mechanism currently resolves [30].

3.2.6 Research and Design Gaps

Federated data governance remains an emerging research area with a limited empirical base. Within the papers from the SLR, gaps within the federated data governance literature has been identified.

To begin with, the literature suggests that individual compliance components have been developed and, in some cases, validated in isolation. Schuiki et al. provide creation-time privacy policy verification at the data product level [44]. Wider et al. demonstrate compositional contract propagation between data products [2]. Wider, Harrer, and Dietz show how large language models can translate evolving regulatory documents into governance decisions without requiring formal rule specification first [3]. However, no paper assembles these components into an end-to-end compliance architecture. The practical problem this creates is visible when a regulation changes. A new privacy requirement, for instance, must propagate through an entire federated architecture. It must update data contracts at every affected output port. It must also trigger the re-verification of existing data products and adjust access decisions across all domains. At the same time, this process must occur without requiring a central team to intervene manually in each domain. The absence of an integrated

architecture means this propagation path is not defined. Organizations implementing federated governance therefore have validated individual building blocks but no design for how those blocks connect into a system that can respond coherently to regulatory change.

This is followed by the absence of a governance transition model. Implementing federated data governance requires an organization to move from a centralized structure to a federated one. This transition is not instantaneous. It passes through intermediate states in which authority is partially decentralized, accountability roles are not yet fully defined, and automated enforcement has not yet replaced manual oversight. Without a model that describes how to sequence this transition, organizations have no basis for deciding which governance capabilities to establish first, what constitutes a stable intermediate configuration, or when a domain is ready to take on full ownership responsibility. The risk is that organizations either stall in a costly hybrid state or attempt full decentralization before the necessary organizational conditions are in place. Lemieux et al. show empirically that the hybrid transition state is the most governance-costly configuration, combining the disadvantages of both centralized and decentralized models [29]. However, they do not provide guidance on how to move through it efficiently. Bode et al. [7] recommend iterative adoption and distinguish between organizational archetypes, which provides a useful starting point, but stops short of specifying progression criteria for each stage, minimum viable governance configurations, or sequencing logic for the transition. As a result, practitioners attempting the transition currently have no principled basis for determining how far along the process they are or what to do next.

In addition to uncertainty about the next steps, the absence of a validated incentive alignment mechanism has also been identified. Federated governance depends on domain teams actively maintaining data products, documenting data, enforcing local policies, and participating in federation-wide governance decisions. None of these activities are the primary job of a domain team. If domain teams have no reason to invest in governance quality beyond avoiding penalties for non-compliance, the federated model degrades into a compliance exercise rather than a functioning governance system. The data products will be technically present but poorly maintained, and the governance body will lack the domain engagement needed to make collaborative policy decisions. Fadler and Legner theorize data product ownership as a distinct accountability role, and Van Eijk et al. address the platform conditions that support domain autonomy, but neither resolves why a domain team would invest governance effort beyond the minimum required [20, 48]. Mendonça et al. propose cooperative principles and fairness mechanisms, but these were developed for multi-organization cooperatives and have not been validated in single commercial organizations where competitive dynamics between domains are different [19]. No paper provides a tested mechanism for aligning domain incentives with governance participation quality inside a single enterprise.

Furthermore, no metrics currently exist to assess the effectiveness of a governance framework. A governance framework that cannot be evaluated cannot be improved, compared with alternatives, or validated as fit for purpose. Without defined metrics, an organization implementing federated governance has no objective basis for determining whether its governance is functioning as intended, where it is failing, or whether one design choice performs better than another. This also means that the field cannot accumulate comparative knowledge across implementations. Bode et al. report improvements in data accessibility, trust, and reduced redundancy, but these are based on practitioner perceptions gathered in interviews rather than measurements taken against predefined criteria [7]. They do not generalize to other organizations or allow comparison between governance configurations. Dolhopolov et al. discuss federated data governance implementation but similarly do not define a replicable evaluation approach [17]. None of the collected papers specifies what a well-functioning federated governance framework should measurably produce, how to assess

whether those outcomes are being achieved, or how to compare governance effectiveness across organizations with different structures or maturity levels.

Moreover, data governance is disconnected from enterprise architecture and IT governance. There was no paper found that explicitly integrates a federated data governance framework with established EA frameworks such as TOGAF or Zachman, nor with IT governance standards such as COBIT or ISO/IEC 38500 [5, 12, 24, 46]. McCullough and Bode et al. acknowledge this disconnect. However, they do not specify how integration should occur, how authority should relate to existing governance bodies, or how duplication of accountability can be avoided.[7, 35]. This absence creates three practical problems. First, it is unclear which governance body holds final authority when federated data governance rules conflict with enterprise architecture policies. Second, domain teams may face competing accountability obligations from the federated council and from the existing IT governance body, with no mechanism for resolution. Third, without a defined integration point, the framework cannot be formally adopted at the enterprise level, because organizations typically require new governance arrangements to fit within their established operating models before approving them institutionally. McCullough defines enterprise information governance as a target operating model and business architecture, which implies the need for alignment with broader enterprise architecture contexts, but does not specify integration points with COBIT or TOGAF [35]. Bode et al. recommend a cross-domain steering unit without specifying its authority relationship to existing IT governance bodies [7].

Finally, there is a persistent ambiguity in the literature which concerns the formal constitution of the central governance function within federated data governance frameworks. The federated model is consistently described as distributing operational governance to domains. At the same time, however, it assumes a central body responsible for setting global standards, managing the self-serve platform, and enforcing interoperability. Despite this central role, the body itself is rarely defined with the same precision applied to domain-level governance. Dehghani characterizes the central function in terms of what it produces, such as global policies, interoperability standards, and platform capabilities. However, he does not specify its authority basis, accountability structure, or organizational positioning. [16]. Bode et al. recommend a cross-domain steering unit as an empirically derived intervention [7]. However, its composition, mandate boundaries, and relationship to existing IT governance bodies remain unspecified. Wider et al. conceptualize the central function primarily through platform logic and automated policy enforcement [2]. In doing so, they effectively substitute technical infrastructure for an organizational definition. As a result, the central governance function is present in every federated framework because its standards-setting and enforcement role is structurally indispensable. Yet it remains formally undefined. It is described in terms of what it does rather than what it is. This leaves unresolved questions about how its authority is legitimated, how it is held accountable, and how it relates to, or differs from, the centralized governance model it is intended to supersede.

4 Research Proposal

The systematic literature review conducted in Sections 2 and 3 establishes both the conceptual landscape of federated data governance and the specific gaps that remain unresolved in the existing literature. This chapter translates that analysis into a structured research proposal. Section 4.1 formulates the research questions that guide the study. Section 4.2 presents the research methodology, including the selected research approach, the scope of the inquiry, the nature of the artifact to be

designed, the strategy for data collection, and the methods by which that data will be analyzed and used to support design decisions.

4.1 Research Questions

RQ: How should the central data governance function be constituted, with respect to organizational positioning and task allocation, to sustain governance coherence and organizational legitimacy in organizations transitioning from centralized to federated data governance?

This question is split into three sub research questions that address its design dimensions: SRQ1: What tasks and decision rights are retained by the central governance function in a federated data governance model, and which should be delegated to domain-level governance units? SRQ2: How should the authority structure and accountability of the central governance function be configured across the transition from a centralized to a federated governance state? SRQ3: What design requirements must a federated data governance framework satisfy to be organizationally viable in large and organizationally complex single-organization contexts?

To explain the sub research questions, SRQ1 addresses the functional boundary of the central governance function. The federated model's coherence depends on a principled division of authority rather than an ad hoc one, and the literature currently does not specify which governance responsibilities are constitutively central and which are domain-appropriate. SRQ2 addresses the temporal dimension of the design problem. Because organizations pass through intermediate configurations in which authority is only partially decentralized and accountability roles are incompletely defined, the framework must function during the transition period, not only in a mature steady state. The hybrid transition state is particularly costly in governance terms, as Lemieux et al. demonstrate empirically, yet no existing framework specifies how to navigate it [29]. SRQ3 addresses the organizational viability conditions of the framework. A governance design that is technically coherent but incompatible with the structural and cultural realities of its target context will not be adopted, and adoption failure is precisely the pattern that the existing literature documents most consistently [7, 15].

4.2 Research Methodology

This research follows the Design Science Research (DSR) methodology as formulated by Wieringa (2014) [53]. The core of DSR creates and evaluates artifacts, which can be constructs, models and methods that addresses organizational problems. The primary output of this research is a governance framework artifact. This is a structured specification of the central governance function in the federated data governance within a single organization. The artifact is constructed from the Systematic Literature Research conducted before and evaluated against the judgment of domain experts. The design approach for artifacts proposed by Wieringa follows the following design cycle:

1. Problem Investigation

The problem analysis is conducted during the literature review in section 2.1.

2. Treatment Design

The artifact will be designed in this part of the cycle

3. Treatment Validation

This research will validate the artifact by conducting expert interviews alongside an implementation

4. Treatment Implementation

This phase analyzes the effect of the treatment, this is excluded for the research

This process can be seen in figure 4. Note that the treatment implementation phase for the artifact is excluded in this figure, as this is outside of the scope of the research.

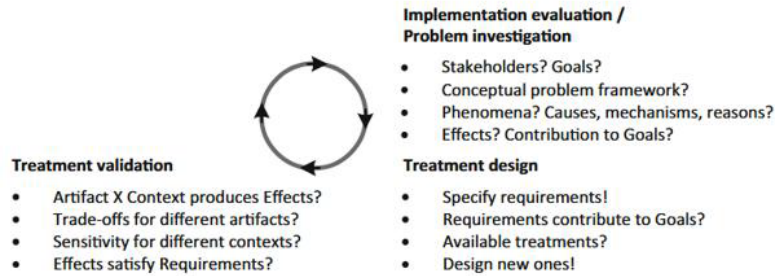


Figure 4: Design Science Engineering Cycle [53]

The artifact to be designed by this research is a conceptual framework. To formulate the design problem, the template from Wieringa can be used [53]:

Improve <a problem context>
By <(re)designing an artifact>
That satisfies <some requirements>
In order to <help stakeholders achieve some goal>

This template is used to clarify the problem context, formulate the requirements and the goals it aims to achieve. This research can therefore be formulated as the following:

Improve the effectiveness and organizational legitimacy of the central data governance function in organizations transitioning toward intra-organizational federated data governance structures

By designing a conceptual framework that specifies the role, authority, and tasks of the central data governance function across the transition to and steady state

That is internally consistent in its allocation of responsibilities, suited to large and organizationally complex single-organization contexts

In order to provide central data governance teams with a clear organizational mandate and actionable guidance on how to govern a federated data structure

4.2.1 Scope

The research concerns single-organization contexts in which federated governance is introduced as an internal design decision. Organizations whose governance is shaped primarily by inter-organizational data-sharing agreements, regulatory requirements for external data exchange, or cross-institutional data spaces fall outside the scope of this study. The authority and accountability structures in those settings differ structurally from those found in intra-organizational federated governance. The framework is intended for large and organizationally complex enterprises in which multiple business domains produce and consume data independently. It is particularly relevant in settings where governance authority remains concentrated in a central function that is structurally distant

from domain-level data production. Organizations operating in a single-domain or small-team configuration are similarly excluded, as the structural rationale for federation does not apply at that scale. The framework does not specify technical platform configurations, data product standards, or enforcement tooling. It is an organizational design artifact concerned with authority structures, accountability relationships, governance roles, and transition sequencing.

4.2.2 Artifact

Following Wieringa's classification, the primary output of this research is a conceptual framework, defined as a structured specification of constructs, relationships, and design principles that prescribes how a problem should be approached [53]. The framework specifies the role, authority, task allocation, and accountability structure of the central governance function across two organizational states: the transition state, in which authority is being progressively redistributed to domain-level units, and the steady state, in which domain-level governance is fully operational. The distinction between these two states is analytically necessary because the appropriate configuration of the central function differs across them, and conflating them is a source of the practical inconsistency documented in the literature [7, 29].

4.2.3 Data Collection

The design and validation of the framework artifact builds on semi-structured expert interviews as its primary data source. Expert interviews are appropriate in this context because the design problem is organizational rather than technical. In addition, the existing literature does not provide sufficient empirical grounding for the specific design decisions the framework must address, particularly with regard to authority, legitimacy, and transition management. Within the research cycle, the interviews serve two distinct and sequential functions. During the design phase, the interviews are intended to generate practitioner knowledge about the current operation of the central governance function. This includes how authority is exercised in practice, how the central function relates to domain-level governance, and what structural challenges emerge during governance transitions. This empirical input is used to develop and refine the framework's constructs and design principles in parallel with the theoretical grounding provided by the literature. During the validation phase, interviews serve as an expert evaluation mechanism. Participants receive the framework artifact in advance of the interview and assess it against four criteria:

1. The consistency of the authority allocation
2. The organizational plausibility of the framework in the target context
3. The completeness relative to the gaps identified in Section 3.2
4. The practical feasibility of its guidance for governance teams

Participant responses inform targeted revisions to the artifact, which are documented and justified with reference to specific evaluative feedback.

4.2.4 Participant Selection

Expert interviews serve two sequential functions in this research:

1. Generating practitioner knowledge to inform framework design

2. Evaluating the framework artifact in the validation phase

Participants are selected using purposive sampling. This is the appropriate strategy when the goal is to generate knowledge from individuals with specific and relevant expertise [11]. Two criteria define eligibility. First, participants must hold or have held a senior role with direct responsibility for data governance, data architecture, or data strategy within a large and organizationally complex enterprise. Second, participants must have direct experience with the design, implementation, or operation of federated or domain-driven governance structures, or with the organizational challenges of transitioning away from a centralized governance model. Practitioners from consulting roles are eligible where they can demonstrate sustained engagement with the governance design of specific organizations.

4.2.5 Data Analysis

Interview data from the design phase is analyzed using thematic analysis [11]. Transcripts are coded inductively in the first pass to identify recurring patterns in how participants describe the central governance function's role, authority, and challenges. In the second pass, codes are organized into themes that correspond to the sub-questions defined in Section 4.1. These themes are then used to inform, challenge, or refine the framework's constructs and design principles. The analysis is conducted in parallel with framework design rather than as a sequential post-processing step, consistent with the iterative character of the DSR cycle.

Data from the validation phase is analyzed through structured qualitative content analysis. Participant responses are mapped against the framework's constructs and evaluated according to the four criteria described above. Revisions to the framework are documented with explicit traceability to the evaluative feedback that motivated them, ensuring methodological transparency and enabling an audit of the design decisions made across iterations.

4.3 Interview Protocol

The interviews will be conducted with individuals representing their respective organizations, drawing on their professional experience and domain expertise. The information gathered may therefore include sensitive organizational or personal information. Accordingly, all ethical requirements of the University of Twente and the applicable faculty's Ethical Committee must be met prior to data collection. This protocol has been submitted for review and approval before any interviews are conducted. The study aims to include approximately 10 interviewees, with 2 to 3 participating in the generative phase and the remainder in the validation phase.

The following precautions are established to ensure full ethical compliance throughout the study:

1. Participants are informed of the study's objectives, scope, and their role prior to the interview, and are given the opportunity to ask clarifying questions before consenting to participate
2. Participation is entirely voluntary. Participants may decline to answer any question and may withdraw from the interview at any time and for any reason, without consequence
3. Informed consent is obtained in writing from each participant before the interview commences, confirming their understanding of and agreement to the conditions set out in this protocol
4. The interview will be conducted in a private setting to ensure confidentiality and to allow participants to speak freely

5. With the participant's explicit consent, the interview will be audio recorded solely for the purpose of transcription. Recordings are deleted upon completion of the transcript
6. Transcripts are anonymized: no personally identifiable information (PII), such as names, job titles, or organizational identifiers, is retained in the written record unless the participant has explicitly consented to being identified
7. Access to raw data, recordings, and transcripts is restricted to the primary researcher. Data will not be shared with third parties
8. All data is used exclusively for academic research purposes
9. Anonymized transcripts may be retained for potential future academic use in accordance with the University of Twente's data retention policy, provided no PII is included

4.3.1 Interview Questions

The interview questions are designed for a 30 minute interview due to the time limitations of the interviewees. Interview questions will be discussed in more depth for interviewees with more time. Below you can find the interview questions for generative and validation purposes. The interview questions are distinguished between generation track A and validation track B

Opening A & B

Q1 Could you briefly describe your current role and your involvement with data governance?

- Is data governance a primary responsibility or one of several?
- How would you characterize the governance model your organization currently operates?

Track A

A2 In your experience, what are the one or two things that only the central data governance function can do that domain teams genuinely cannot do themselves?

A3 When the central function sets a policy or standard, how is that received in practice? Where does it hold, and where does it break down?

- What happens when domain practice conflicts with central policy, and who resolves it?
When extra time is available
- How did the role of the central governance function change during the transition? Did its authority increase, decrease, or transform in character?

A4 If you were advising a central governance team entering a federated transformation, what would you tell them their actual job is, beyond the official mandate?

- Where do you most often see the central function misjudge its own role or overreach?

Track B

B2 Having read the framework, does the authority structure assigned to the central governance function match what you would expect in a functioning federated organization? Which specific authority claim do you view as least defensible, or most context-dependent?

- Is anything overclaimed, underclaimed, or missing from the central function's mandate?

B3 The framework describes a transition sequence with identifiable stages. Does that sequence reflect how transitions actually unfold in your experience?

- Is there a stage the framework underestimates in terms of difficulty or duration?
- Is there a stage that in your experience is frequently skipped, and what are the consequences?

B4 If you were implementing this framework in your organization, what would you change, and what would you keep exactly as it is?

Closing A & B

Q5 Is there something important about how the central governance function operates in federated organizations that I have not asked about, or that you think is frequently misunderstood?

4.4 Planning

4.4.1 Design Proposition

The central claim guiding this research is that the operational failures observed in organizations transitioning toward federated data governance are not primarily caused by the absence of technical tooling, but by the absence of a principled specification of the central governance function's authority, tasks, and accountability structure across the transition. From this diagnosis, the following design proposition is drafted: A conceptual framework resulting in a viable governance design artifact for large and organizationally complex enterprises. The framework explicitly specifies the authority boundaries, task allocation, and accountability structure of the central governance function across both the transition state and the federated steady state. Its viability depends on several conditions. First, it must be internally consistent in how it allocates responsibilities. Second, it must reduce the structural ambiguity documented in the existing literature with respect to the mandate of the central function. Third, it must provide design guidance that is sufficiently actionable for practitioners navigating the transition. In doing so, it should help organizations avoid reverting to centralized control or remaining in an unresolved hybrid state.

The claim will be assessed against the four validation criteria defined in Section 4.2.3: authority consistency, organizational plausibility, completeness relative to the identified gaps, and practical feasibility.

4.4.2 Research Procedure

The research is organized into four sequential phases that correspond to the DSR cycle described in Section 4.2.

Phase 1 covers literature synthesis and initial framework design. The systematic literature review conducted in Sections 2 and 3 provides the conceptual and empirical grounding for the framework. Drawing on the thematic synthesis, and in particular on the gaps identified in Section 3.2.6, an initial framework specification is produced. This specification defines the authority boundaries, task allocation, and accountability structure of the central governance function across the transition state and the federated steady state.

Phase 2 covers framework consolidation and preparation for validation. The initial specification is reviewed for internal consistency and alignment with the design requirements established in Section 4.2.1. Interview instruments are finalized and ethical approval is confirmed. Participants are recruited and scheduled in accordance with the selection criteria defined in Section 4.2.4. The framework artifact is prepared in a form suitable for presentation to participants ahead of the validation interviews.

Phase 3 covers validation and final revision. Design-phase interviews are conducted first to generate practitioner input that may refine the framework before formal evaluation. Validation-phase interviews follow, in which participants assess the artifact against the four criteria defined in Section 4.2.3. Interview data is analyzed using the methods described in Section 4.2.5. Revisions to the framework are made with explicit traceability to evaluative feedback, and the final artifact is produced.

Phase 4 covers thesis writing and submission. The complete thesis integrates the literature review, framework design, validation findings, and discussion of limitations and possible generalization. Final revisions are made following supervisor feedback before submission.

4.4.3 Research Schedule

The research is planned across a period of approximately twenty weeks, divided in accordance with the four phases described in Section 4.4.2. The schedule below represents a target distribution. Phases 1 and 2 overlap partially, as framework design is initiated in parallel with the later stages of the literature review.

Phase	Activity	Indicative Duration
1	Literature synthesis and initial framework design	Weeks 1–6
2	Framework consolidation, ethical approval, participant recruitment	Weeks 5–10
3	Interviews, analysis, framework revision	Weeks 10–16
4	Thesis writing and submission	Weeks 14–20

Phases 3 and 4 overlap from week 14 onward, as thesis writing commences while final validation revisions are being completed. Ethical approval is treated as a hard dependency for Phase 3. No interview activity begins before approval is confirmed.

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A Appendix A

Table 4: Quality assessment scores for all 40 included papers. Scores: 1 = fully met; 0.5 = partially met; 0 = not met. Type codes: *J* = peer-reviewed journal; *C* = conference paper; *B* = book/monograph; *Ch* = book chapter; *T* = tutorial/short paper. Criteria: QA1 = Research question clear; QA2 = Theoretical grounding; QA3 = Methodology described; QA4 = Data/evidence identified; QA5 = Findings traceable; QA6 = Limitations acknowledged; QA7 = Governance structures; QA8 = Context described; QA9 = Core constructs in practice; QA10 = Generalisable conclusions. ** bib key not in `references.bib` – add manually. * bib key estimated – verify against your `.bib` file.

#	Paper / Authors / Year	Type	QA1	QA2	QA3	QA4	QA5	QA6	QA7	QA8	QA9	QA10	Total
1	AI-Assisted Data Governance with Data Mesh Manager [3]	<i>C</i>	1	0.5	0.5	0	0.5	0.5	1	1	0.5	0.5	6/10
2	Decentralized IG Framework – Blockchain in Healthcare [37]	<i>C</i>	1	1	1	1	1	0.5	1	1	1	1	9.5/10
3	A Framework for Operationalizing Cooperative Principles in Data Governance [19]	<i>C</i>	1	1	0.5	1	1	0.5	1	1	1	1	9/10
4	A Polycentric Governance Lens on Data Infrastructures [40]	<i>J</i>	1	1	1	1	1	0.5	1	1	1	1	9.5/10
5	Decentralized Data Governance as Part of a Data Mesh Platform [52]	<i>C</i>	1	0.5	0.5	0.5	1	0.5	1	1	0.5	0.5	7/10
6	A More Inclusive Europe through Personal Data Sovereignty in Cross-Border Digital Public Services [45]	<i>C</i>	0.5	0.5	0	0.5	0.5	0	1	1	0.5	0.5	5/10
7	Adopting Data Mesh Principles to Boost Data Sharing for Clinical Trials [33]	<i>C</i>	1	0.5	0.5	0	0.5	0	1	1	0.5	0.5	5.5/10
8	Architectural Design Decisions for Self-Serve Data Platforms in Data Meshes [48]	<i>C</i>	1	0.5	1	1	1	0.5	0.5	1	0.5	1	8/10
9	Blockchain-Aided and Privacy-Preserving Data Governance in Multi-Stakeholder Applications [42]	<i>J</i>	1	1	1	1	1	0.5	1	1	1	1	9.5/10
10	Blockchain-Enabled Data Governance for Privacy-Preserving Sharing of Confidential Data [55]	<i>J</i>	1	1	1	1	1	0.5	0.5	1	0	0.5	7.5/10
11	Toward Avoiding the Data Mess: Industry Insights from Data Mesh Implementations [7]	<i>J</i>	1	1	1	1	1	1	1	1	1	1	10/10

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#	Paper / Authors / Year	Type	QA1	QA2	QA3	QA4	QA5	QA6	QA7	QA8	QA9	QA10	Total
12	Caught in the Middle: Blockchain and Information Governance [29]	<i>J</i>	1	1	1	1	1	0.5	1	1	1	1	9.5/10
13	Designing Data Governance Mechanisms for Data Marketplace Metaplatfroms	<i>C</i>	1	1	0.5	0.5	0	0	0.5	1	0.5	1	6/10
14	Data Mesh Architecture: From Theory to Practice [28]	<i>T</i>	0.5	1	0	0.5	0.5	0	0.5	1	0.5	0.5	5/10
15	Data Mesh: Guiding Principles and Patterns, and Data Catalog Architectural Concept [25]	<i>C</i>	0.5	0.5	0	0	0.5	0	0.5	0.5	0.5	0.5	3.5/10
16	Data Mesh: Principles, Patterns, Architecture [36]	<i>B</i>	1	1	0.5	0	0.5	0	1	1	1	1	7/10
17	Data Mesh in Action [34]	<i>B</i>	1	1	0.5	0.5	0.5	0	0.5	1	0.5	1	6.5/10
18	Data Spaces: Design, Deployment and Future Directions [13]	<i>B</i>	0.5	1	0.5	0.5	0.5	0.5	0.5	1	0.5	1	6.5/10
19	Data Mesh: Delivering Data-Driven Value at Scale [16]	<i>B</i>	1	1	0.5	0.5	1	0	1	1	1	1	8/10
20	Decentralized Services Computing Paradigm for Blockchain-Based Data Governance: Programmability, Interoperability, and Intelligence [31]	<i>J</i>	1	1	0	0.5	0.5	0.5	1	0.5	0.5	1	6.5/10
21	Decentralized Management Model for Personal Information Governance Oriented Toward Data Owners [10]	<i>J</i>	1	0.5	1	1	1	0.5	0.5	0.5	1	0.5	7.5/10
22	Defining a New Perspective: Enterprise Information Governance [35]	<i>J</i>	1	1	0.5	1	1	0.5	1	1	1	1	9/10
23	Enabling Trusted Data Sharing in Data Spaces – PROTON: A Privacy-by-Design Approach to Data Products [44]	<i>C</i>	1	1	0.5	0.5	1	0	0.5	1	0.5	0.5	6.5/10
24	Enterprise Data Strategy: A Decentralized Data Mesh Approach [47]	<i>C</i>	0.5	0.5	0	0	0.5	0	0.5	0.5	0.5	0.5	3.5/10
25	Data Ownership Revisited: Clarifying Data Accountabilities in Times of Big Data and Analytics [20]	<i>J</i>	1	1	1	1	1	0.5	1	1	1	1	9.5/10
26	Federated Governance and Decentralized Identities for Participatory Sensing Data Mesh [14]	<i>C</i>	1	0.5	0.5	0	0.5	0	1	0.5	0.5	0.5	5/10
27	From Data Mesh to Intermesh: A Platform-Driven Approach to Govern Inter-Organizational Data Sharing [51]	<i>C</i>	1	1	0.5	0.5	0.5	0.5	1	1	1	1	8/10

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#	Paper / Authors / Year	Type	QA1	QA2	QA3	QA4	QA5	QA6	QA7	QA8	QA9	QA10	Total
28	HealthMesh: Federated Healthcare Data Management [6]	<i>C</i>	1	0.5	0.5	0.5	1	0.5	0.5	1	0.5	1	7/10
29	Implementing a Blockchain-Powered Metadata Catalog in Data Mesh Architecture [18]	<i>C</i>	1	0.5	0.5	0.5	1	0.5	0.5	1	0.5	0.5	6.5/10
30	Internet Interoperability and Polycentric Attributes in Global Digital Data Ordering [4]	<i>Ch</i>	1	1	0.5	0.5	1	0.5	1	1	0.5	1	8/10
31	Trustworthy Data Space: Collaborative Trust Mechanism Driven by Blockchain Technology [30]	<i>J</i>	1	1	0.5	1	1	0.5	1	1	1	1	9/10
32	Platform-Centric Agile Transformation and Federated Data Governance [49]	<i>C</i>	0.5	0.5	0.5	0	0.5	0	1	0.5	0.5	0.5	4.5/10
33	Scalable Policy-as-Code Decision Points for Data Products [32]	<i>C</i>	1	0.5	1	1	1	0.5	0.5	1	0.5	1	8/10
34	Security and Ownership in User-Defined Data Meshes [41]	<i>J</i>	1	0.5	1	0.5	1	0.5	0.5	1	1	0.5	7.5/10
35	Software-Defined Infrastructure for Decentralized Data Lifecycle Governance [23]	<i>C</i>	1	1	0.5	0.5	0.5	1	1	1	1	1	8.5/10
36	TAO-Based Data Governance in Intelligent Transportation Systems [54]	<i>C</i>	1	1	0.5	0.5	0.5	0.5	1	1	1	0.5	7.5/10
37	Towards Automating Federated Data Governance [2]	<i>C</i>	1	1	1	0.5	1	0.5	1	1	0.5	1	8.5/10
38	Understanding Data Spaces: A Systematic Mapping Study of Foundations, Technical Building Blocks, and Sectoral Adoption [27]	<i>J</i>	1	1	1	1	1	1	1	1	1	1	10/10
39	Unlocking the Power of Digital Commons: Data Cooperatives as a Pathway for Data Sovereignty [8]	<i>J</i>	1	1	0.5	0.5	1	0.5	1	1	1	1	8.5/10
40	IT Governance: How Top Performers Manage IT Decision Rights for Superior Results [50]	<i>B</i>	1	1	1	1	1	0.5	1	1	1	1	9.5/10